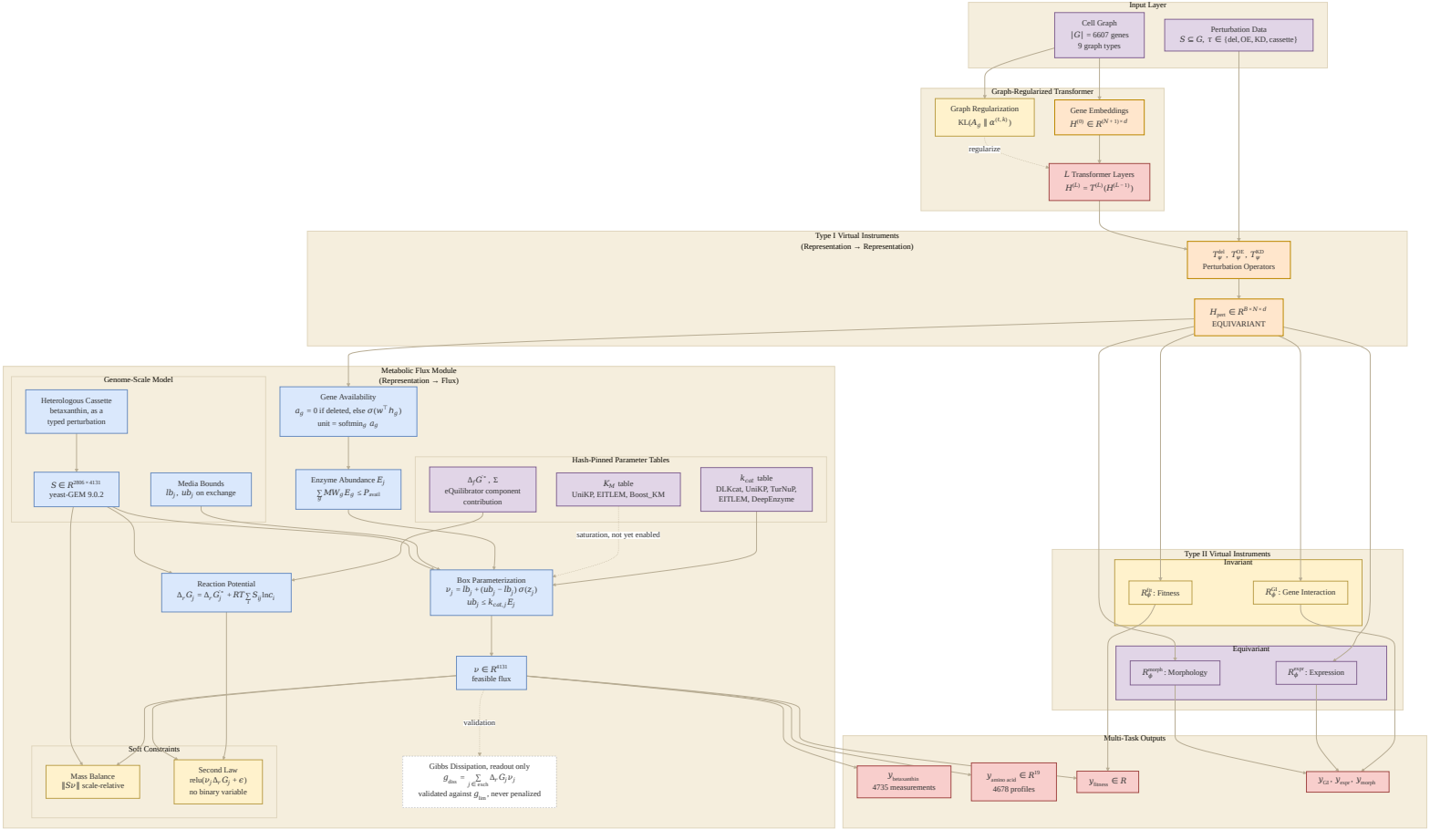

Input Layer
Cell Graph
$|G| = 6607$ genes
9 graph types
Perturbation Data
$S \subseteq G$, $\tau \in$ {del, OE, KD, cassette}
Graph-Regularized Transformer
Graph Regularization
$\mathrm{KL}(A_g \,\|\, a^{(l,h)})$
Gene Embeddings
$H^{(0)} \in R^{(N+1)\times d}$
regularize
L Transformer Layers
$H^{(L)} = T^{(L)}(H^{(L-1)})$
Type I Virtual Instruments
(Representation → Representation)
T_ψ^{del}, T_ψ^{OE}, T_ψ^{KD}
Perturbation Operators
$H_{\mathrm{pert}} \in R^{B\times N\times d}$
EQUIVARIANT
Genome-Scale Model
Heterologous Cassette
betaxanthin, as a
typed perturbation
$S \in R^{2806\times 4131}$
yeast-GEM 9.0.2
Media Bounds
lb_j, ub_j on exchange
Metabolic Flux Module
(Representation → Flux)
Gene Availability
$a_g = 0$ if deleted, else $\sigma(w^\top h_g)$
unit = softmin$_g$ a_g
Enzyme Abundance E_j
$\sum_g MW_g E_g \le P_{\mathrm{avail}}$
Hash-Pinned Parameter Tables
$\Delta_r G'^\circ$, Σ
eQuilibrator component
contribution
K_M table
UniKP, EITLEM, Boost_KM
k_{cat} table
DLKcat, UniKP, TurNuP,
EITLEM, DeepEnzyme
saturation, not yet enabled
Reaction Potential
$\Delta_r G_j = \Delta_r G_j'^\circ + RT\sum_i S_{ij}\ln c_i$
Box Parameterization
$v_j = lb_j + (ub_j - lb_j)\,\sigma(z_j)$
$ub_j \le k_{\mathrm{cat},j} E_j$
$v \in R^{4131}$
feasible flux
validation
Soft Constraints
Mass Balance
$\|Sv\|$ scale-relative
Second Law
$\mathrm{relu}(v_j \Delta_r G_j + \epsilon)$
no binary variable
Gibbs Dissipation, readout only
$g_{\mathrm{diss}} = \sum_{j\in\mathrm{exch}} \Delta_r G_j v_j$
validated against g_{bio}, never penalized
Type II Virtual Instruments
Invariant
R_ϕ^{fit}: Fitness
R_ϕ^{GI}: Gene Interaction
Equivariant
R_ϕ^{morph}: Morphology
R_ϕ^{expr}: Expression
Multi-Task Outputs
$y_{\mathrm{betaxanthin}}$
4735 measurements
$y_{\mathrm{amino\ acid}} \in R^{19}$
4678 profiles
$y_{\mathrm{fitness}} \in R$
y_{GI}, y_{expr}, y_{morph}